

Optical Fiber Communications



**— Chapter 2 —
— Optical fiber waveguides —**

Optical fiber waveguides

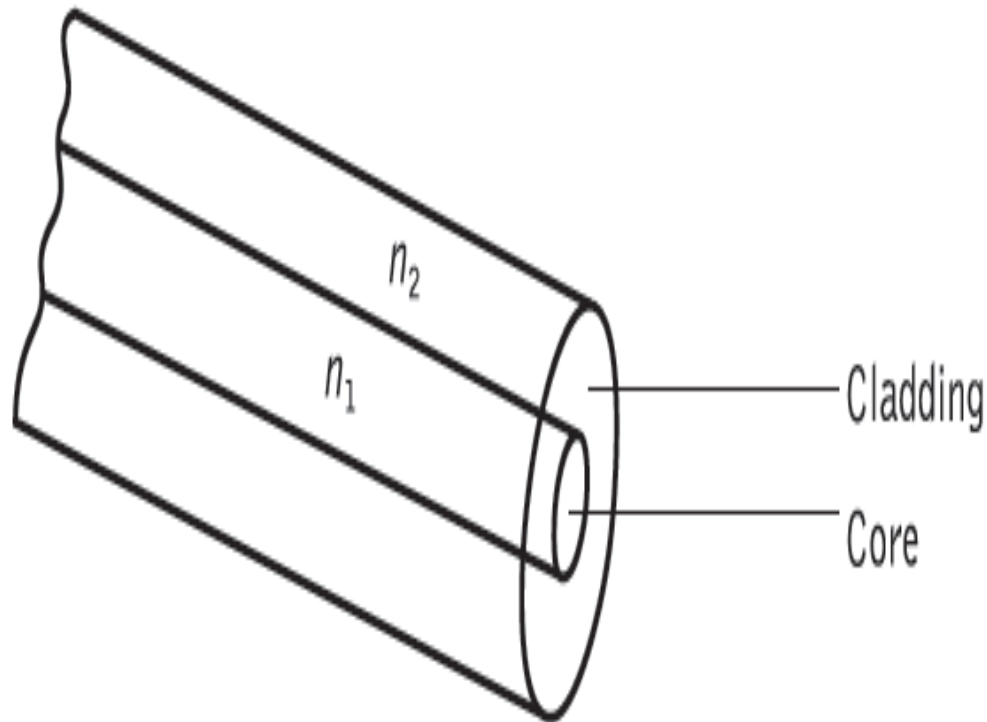


Figure 2.1 Optical fiber waveguide showing the core of refractive index n_1 , surrounded by the cladding of slightly lower refractive index n_2

Total internal reflection

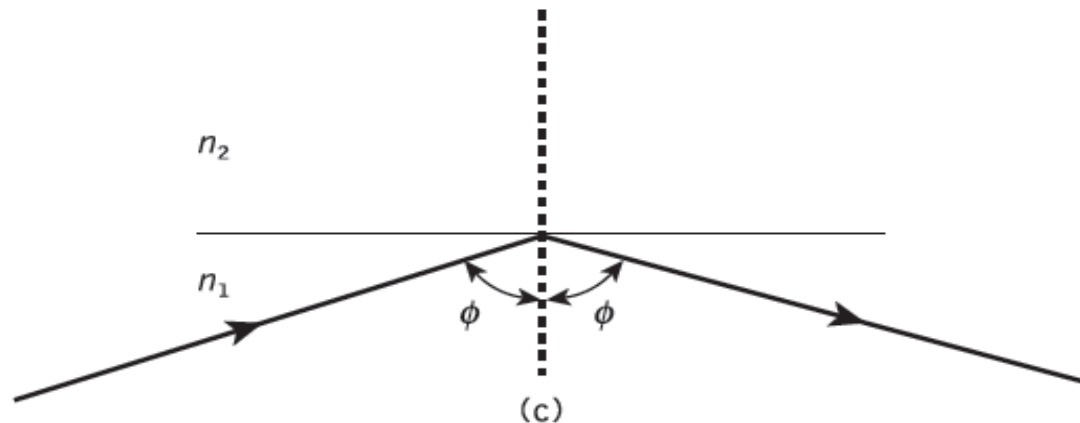
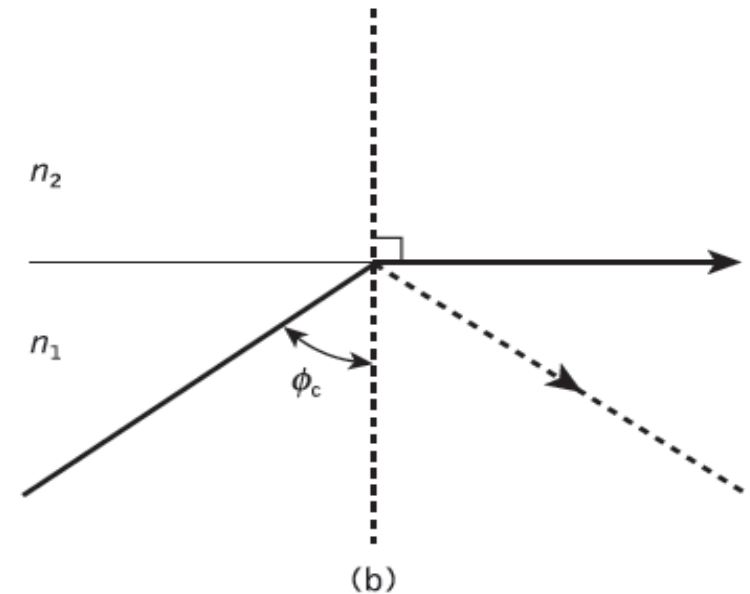
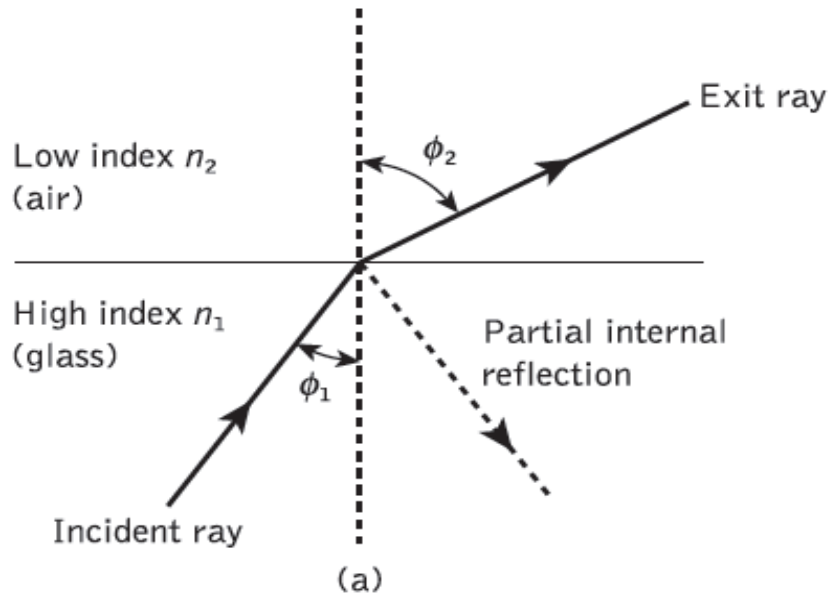
- The angles of incidence ϕ_1 and refraction ϕ_2 are related to each other and to the refractive indices of the dielectrics by Snell's law of refraction which states that

- $n_1 \sin \phi_1 = n_2 \sin \phi_2$

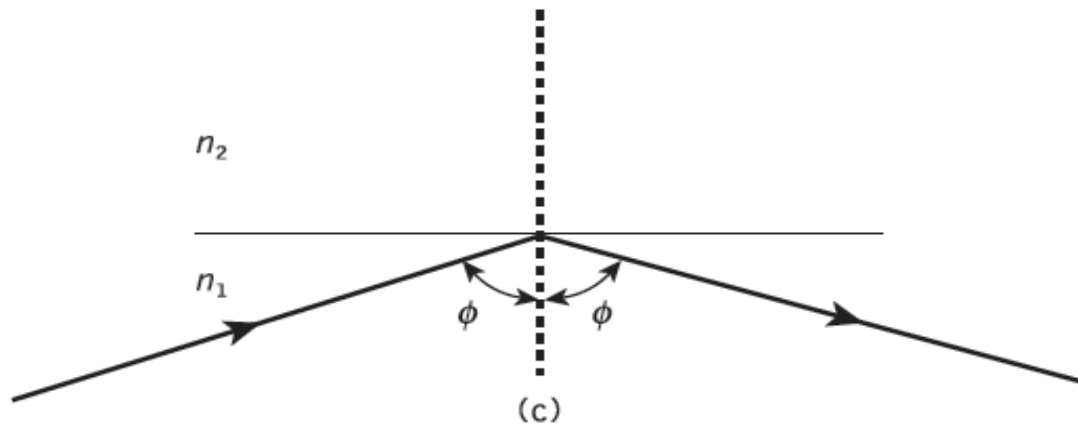
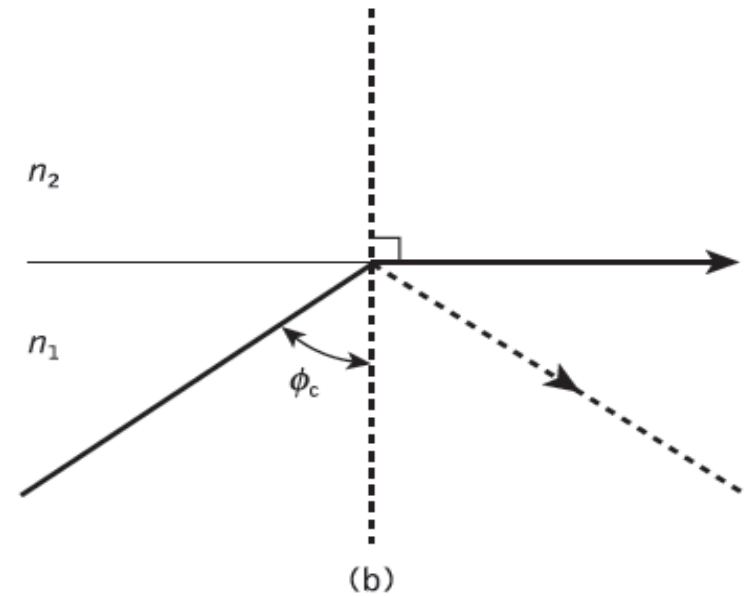
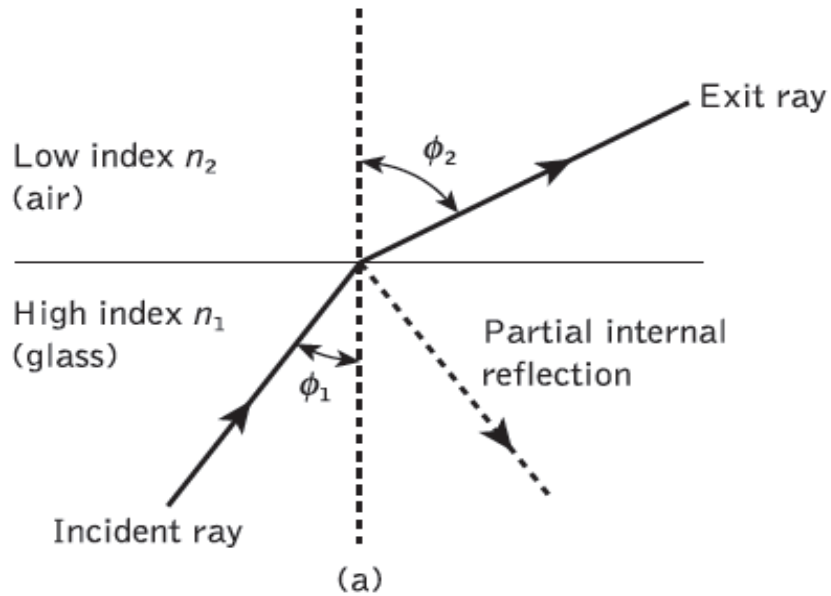
- Or

$$\frac{\sin \phi_1}{\sin \phi_2} = \frac{n_2}{n_1}$$

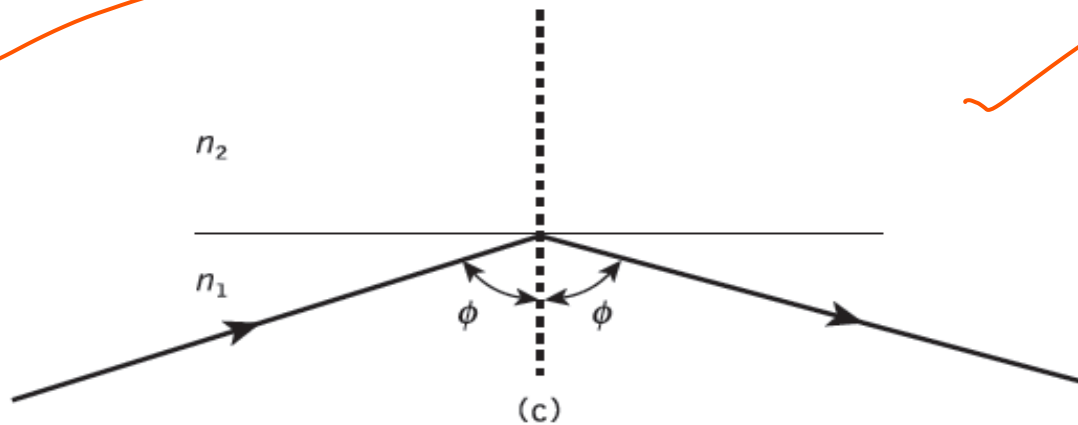
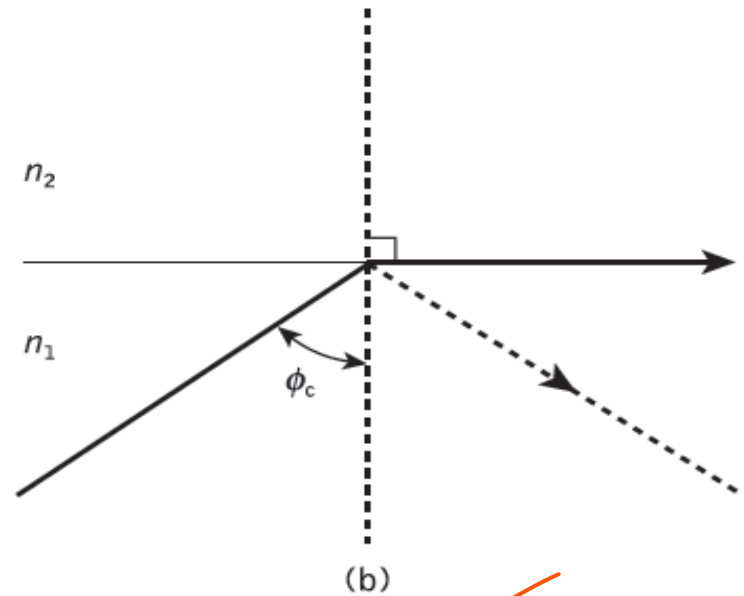
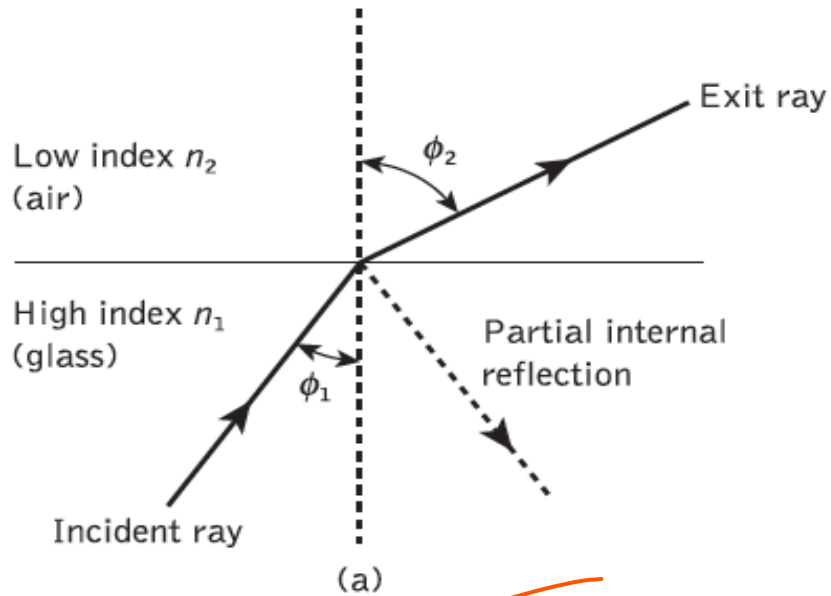
Light rays incident on a high to low refractive index interface (e.g. glass–air): (a) refraction



Light rays incident on a high to low refractive index interface (e.g. glass–air): (b) the limiting case of refraction showing the critical ray at an angle ϕ_c



Light rays incident on a high to low refractive index interface (e.g. glass–air): (c) total internal reflection where $\phi > \phi_c$



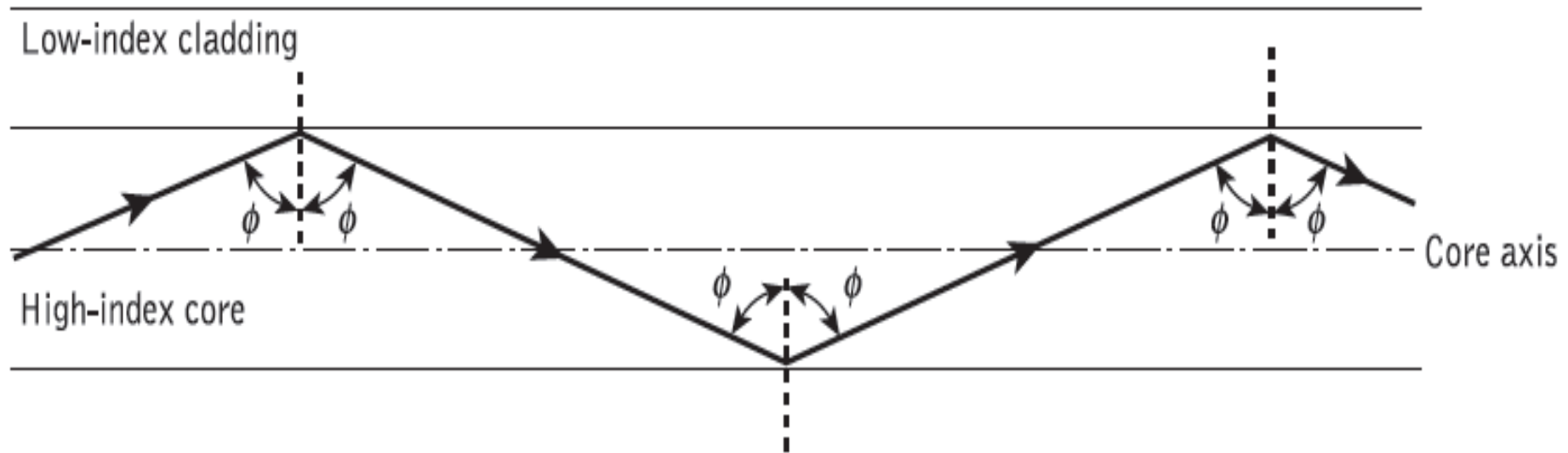
Total internal reflection



- The value of the critical angle is given by

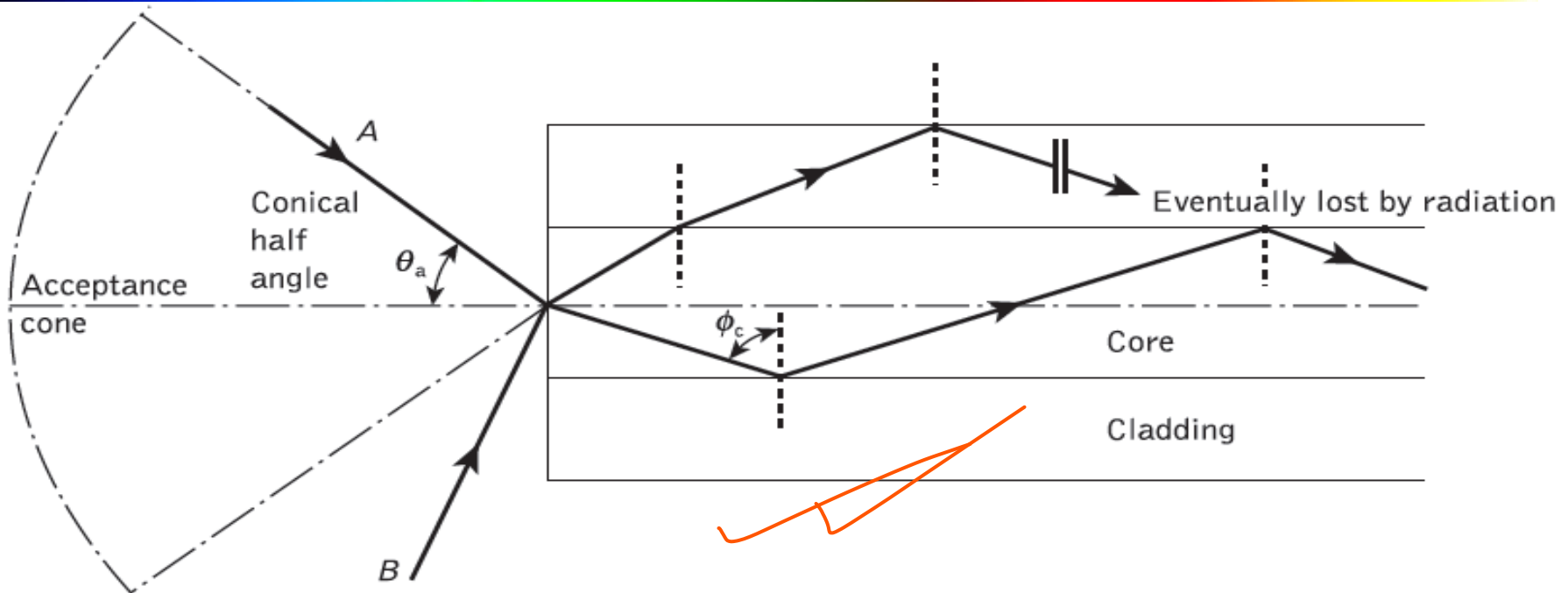
$$\sin \phi_c = \frac{n_2}{n_1}$$

The transmission of a light ray in a perfect optical fiber



At angles of incidence greater than the critical angle the light is reflected back into the originating dielectric medium (total internal reflection) with high efficiency (around 99.9%).

The acceptance angle θ_a when launching light into an optical fiber



Any rays which are incident into the fiber core at an angle greater than θ_a will be transmitted to the core-cladding interface at an angle less than ϕ_c , and will not be totally internally reflected.

θ_a is sometimes referred to as the maximum or total acceptance angle

meridional rays vs Skew rays

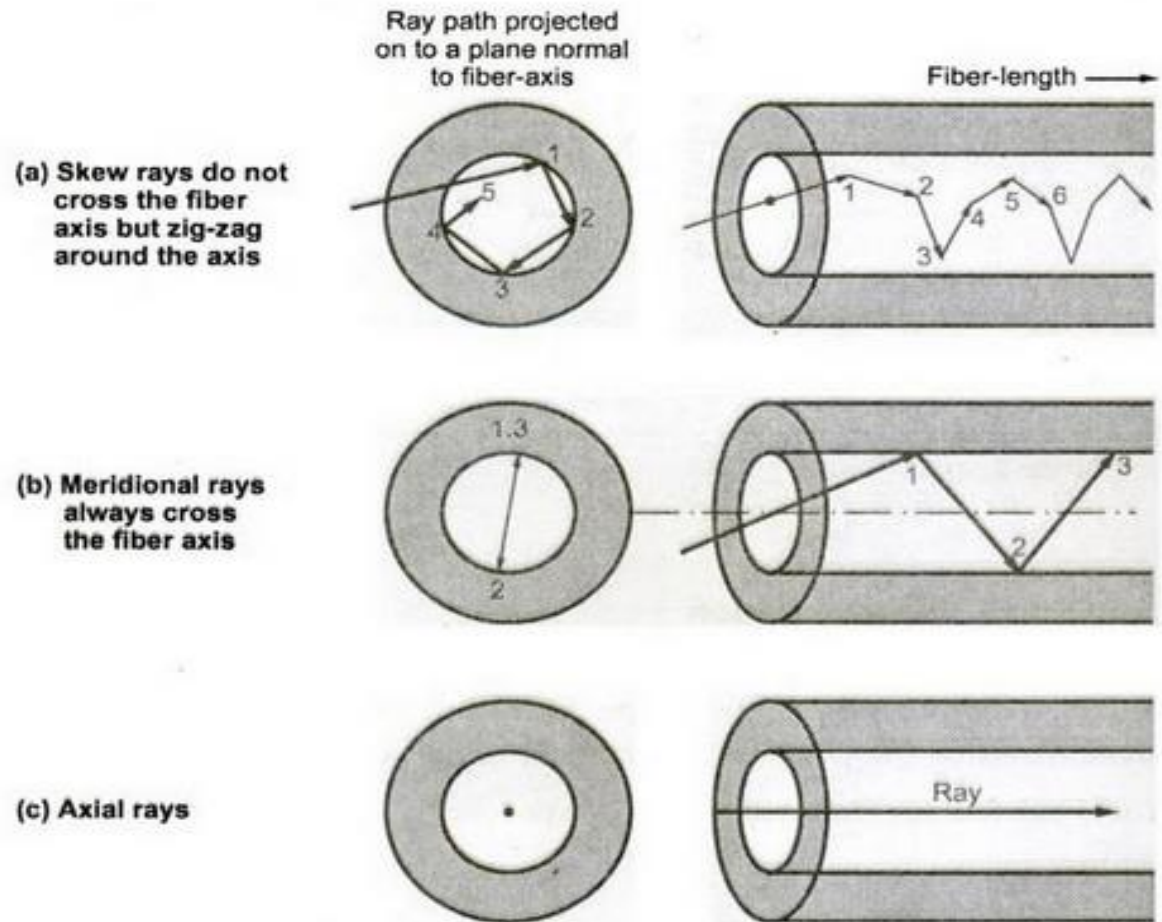


Fig. 11 Different ray propagation

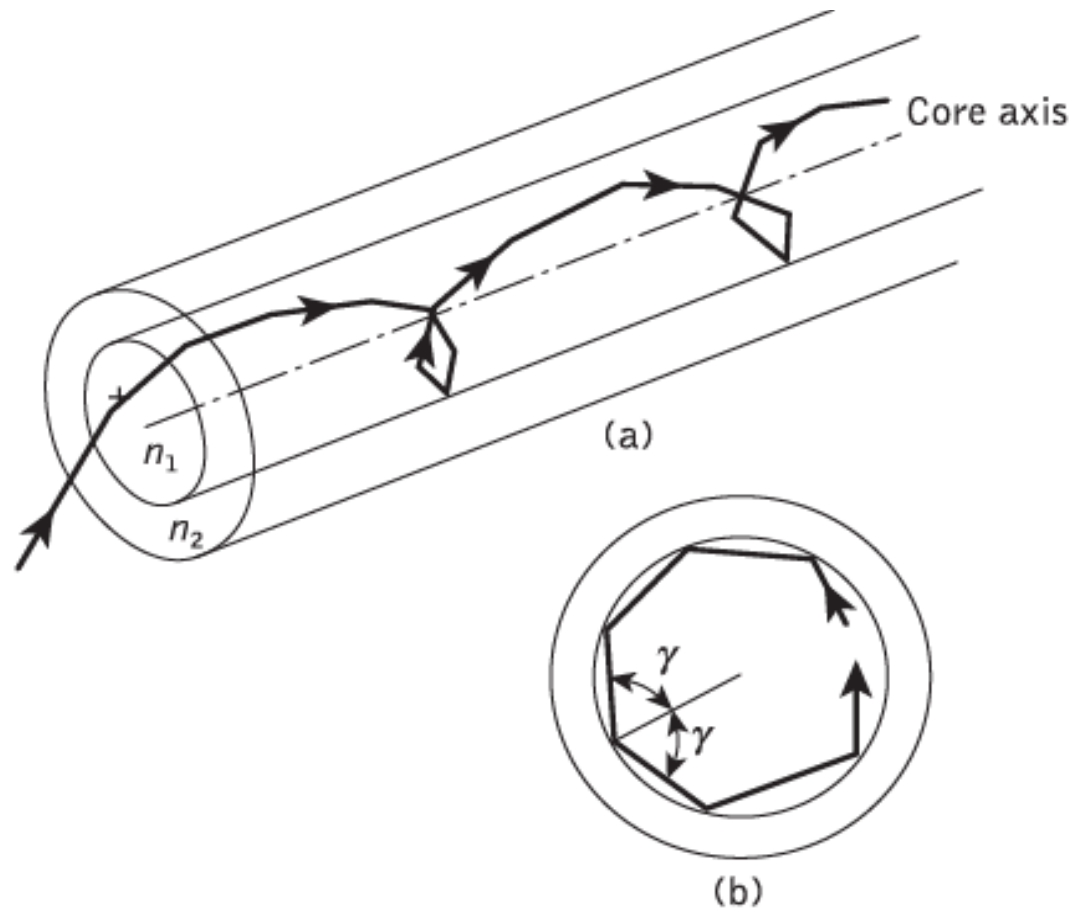
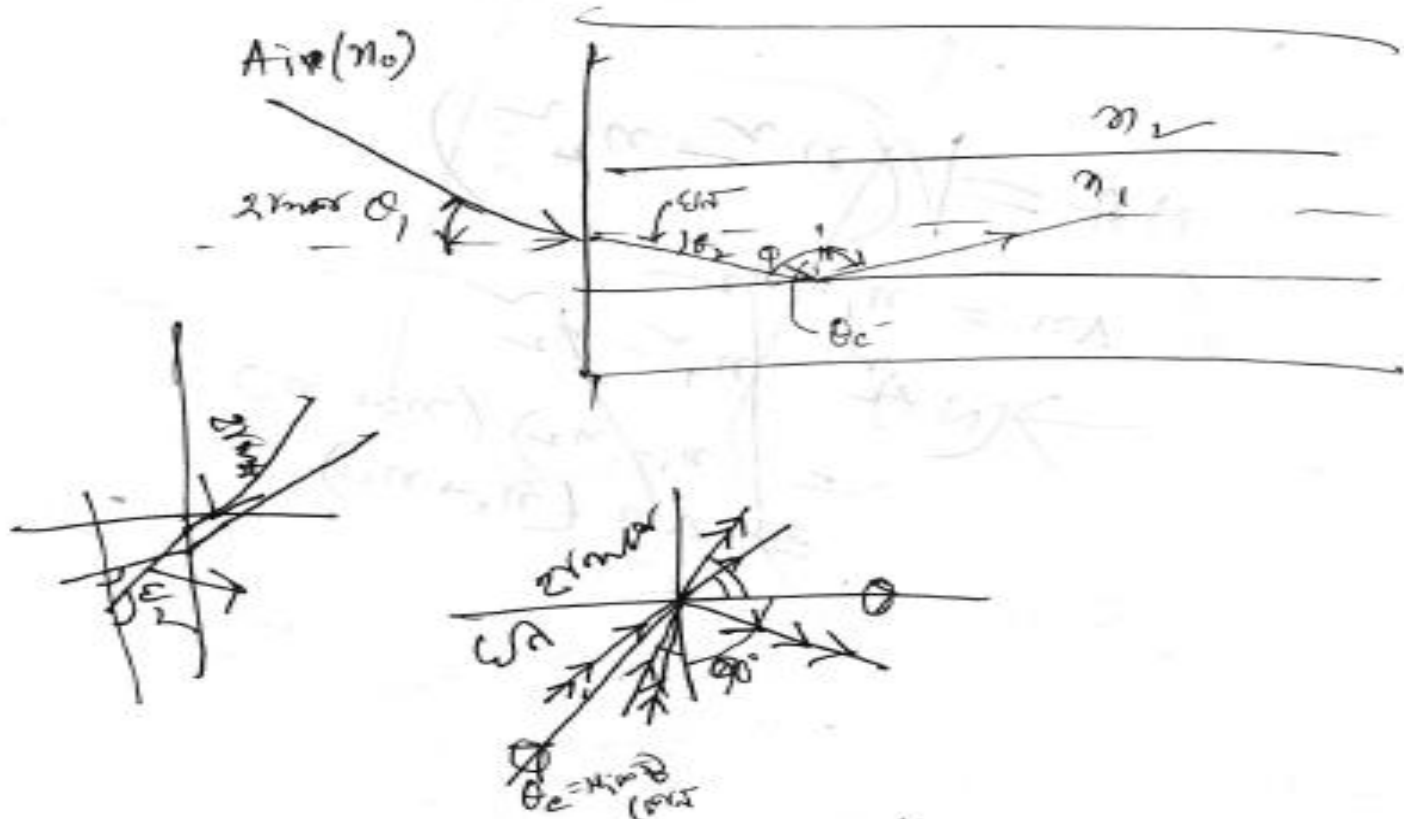


Figure 2.6 The helical path taken by a skew ray in an optical fiber: (a) skew ray path down the fiber; (b) cross-sectional view of the fiber

$$n_0 \sin \theta_1 = n_1 \sin \theta_2$$

$$\phi = \frac{\pi}{2} - \theta_2$$

$$\begin{aligned} n_0 \sin \theta_1 &= n_1 \sin \left(\frac{\pi}{2} - \phi \right) \\ &= n_1 \cos \phi \\ &= n_1 \sqrt{1 - \sin^2 \phi} \end{aligned}$$



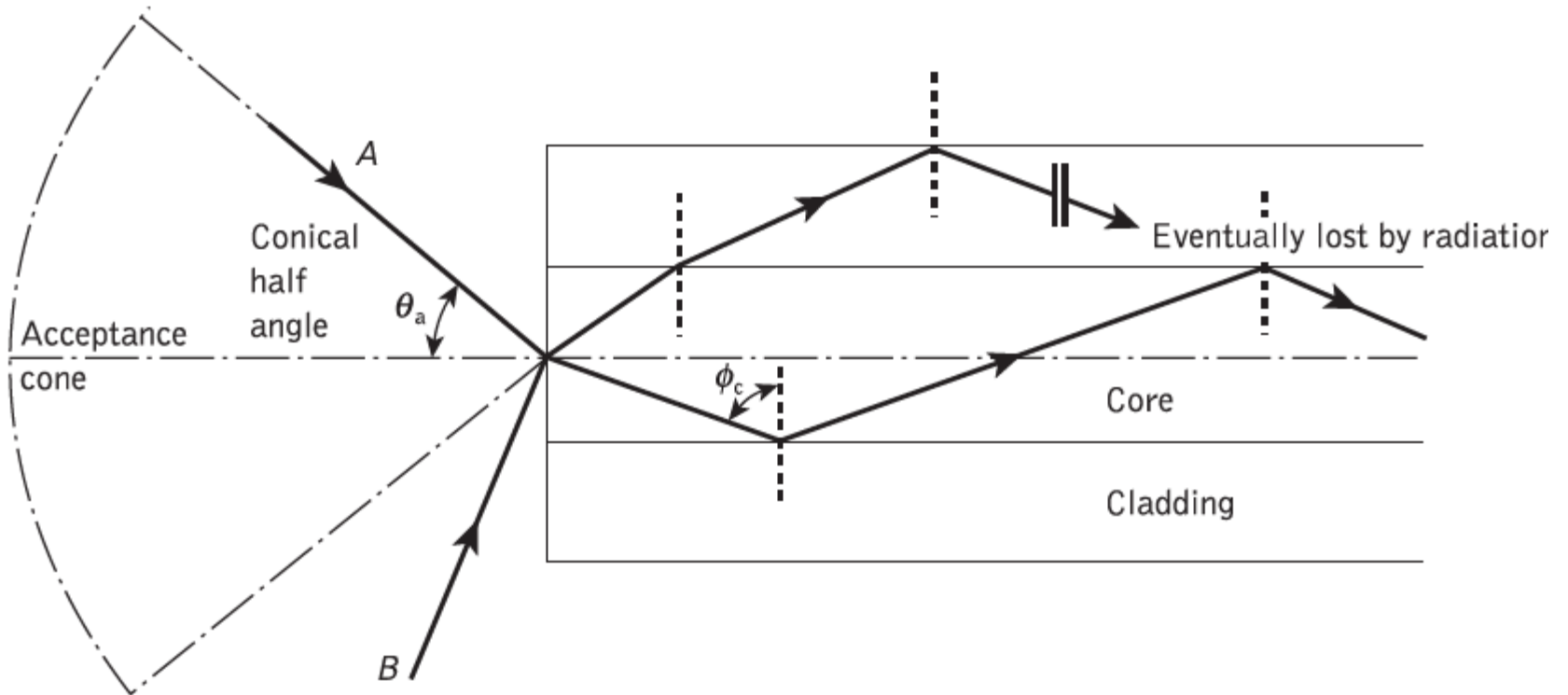


Figure 2.4 The acceptance angle θ_a when launching light into an optical fiber

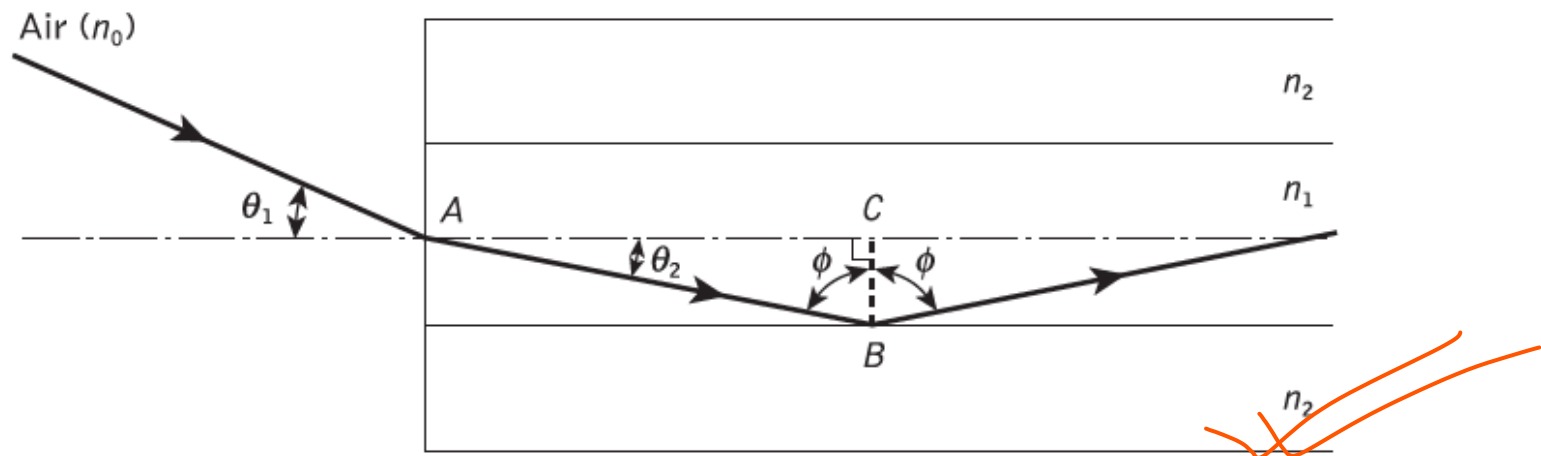
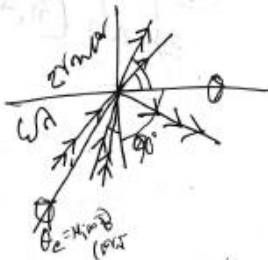
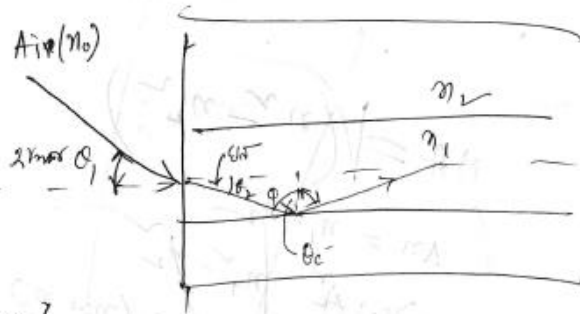


Figure 2.5 The ray path for a meridional ray launched into an optical fiber in air at an input angle less than the acceptance angle for the fiber

$$n_0 \sin \theta_1 = n_1 \sin \theta_2$$

$$\phi = \frac{\pi}{2} - \theta_2$$

$$\begin{aligned} n_0 \sin \theta_1 &= n_1 \sin \left(\frac{\pi}{2} - \phi \right) \\ &= n_1 \cos \phi \\ &= n_1 \sqrt{1 - \sin^2 \phi} \end{aligned}$$



$$\cos 0 = 1$$

$$\sin 90 = 1$$

$$n_1 \sin \theta_c = n_2 \sin 90$$

$$\frac{\sin \theta_c}{\sin 90} = \frac{n_2}{n_1}$$

$$n_0 \sin \theta_1 = n_1 \sqrt{1 - \sin^2 \phi} \quad \text{--- 2.6}$$

$$\Rightarrow n_0 \sin \theta_1 = n_1 \sqrt{1 - \sin^2 \phi}$$

$$= n_1 \sqrt{1 - \frac{n_2^2}{n_1^2}}$$

$$= n_1 \sqrt{\frac{n_1^2 - n_2^2}{n_1^2}}$$

$$= \sqrt{n_1^2 - n_2^2} \quad \text{--- 2.7}$$

NA = numerical aperture is defined as

$$NA = n_0 \sin \theta_a = \sqrt{n_1^2 - n_2^2} \quad \text{--- 2.8}$$

n_0 for air = 1

$$\therefore \sin \theta_a = \sqrt{n_1^2 - n_2^2}$$

incident meridional rays over the range $0 \leq \theta_1 \leq \theta_a$ will be propagated within the fiber.

⇒ The NA may also be given in terms of the relative refractive index difference betⁿ the core & the cladding which is defined as

$$\Delta = \frac{n_1^2 - n_2^2}{2n_1^2}$$

$$\approx \frac{(n_1 - n_2)(n_1 + n_2)}{2n_1^2}$$

$$\approx \frac{(n_1 - n_2)}{2n_1} \quad \text{for } \Delta \ll 1$$

for $\Delta \ll 1$ the cladding refractive index $n_2 \approx n_1$

$$NA = \sqrt{n_1^2 - n_2^2} \approx n_1 \sqrt{2\Delta}$$

$$\textcircled{*} 2\Delta n_1^2 = n_1^2 - n_2^2$$

$$\Rightarrow \sqrt{n_1^2 - n_2^2} = \sqrt{2\Delta n_1^2}$$

$$\Rightarrow NA = n_1 \sqrt{2\Delta} \quad \text{--- GE relationship is}$$

very useful measure of light-collecting ability of a fiber.

They are independent of fiber core diameter & will hold for diameters as small as 8 μm .